Zhang 2019 Follicular Lymphoma — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) integrates all 4 samples + both patients (FL1018/FL2001): per-neighbourhood iLISI 1.05→1.88 of a 3.11 perfect-mixing ceiling. One connected component (0 islands removed). Judged on the regenerated embedding + per-neighbourhood iLISI across a θ sweep (θ=1 archipelago → θ=10 mixed): mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

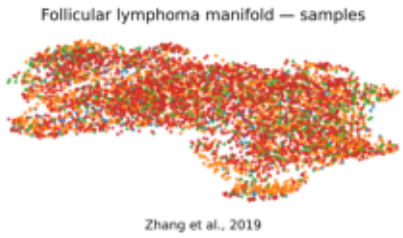
Non-branching case: a single Gast → Crest developmental gradient. Module expression is weak (raw means ~0.01). The Gast→Crest axis is spatially organized and confound-independent: along pseudotime Gast falls (ρ=−0.51, survives regressing out progression+interferon+proliferation) and Crest rises to a terminal gradient (ρ=+0.32). pSom is the exception — ~63% confound-explained, high-decile 54% cycling (vs 14% baseline), so pSom is proliferation-confounded, not an independent axis. Gast↔pSom mild cross-repression (r=−0.18, no hub); Crest opposes both (r=−0.54/−0.60) → inverse terminal gradient, no Gast-arm inversion.

Gate 3 — hub cross-repression → NA

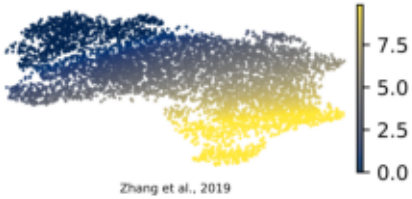
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks robust across PPT Nodes=15/20/30; no Gast/pSom co-expression hub). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

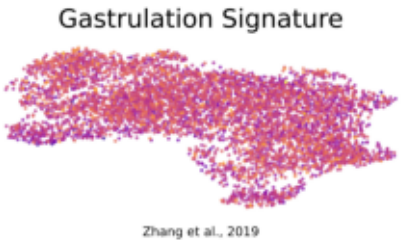
Manifold — samples



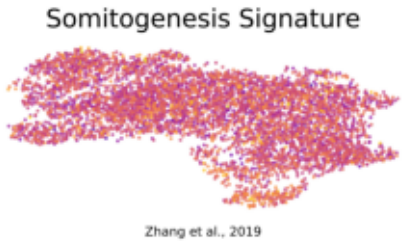
Pseudotime



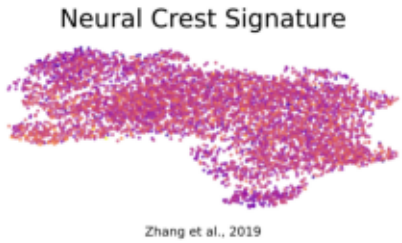
Gastrulation



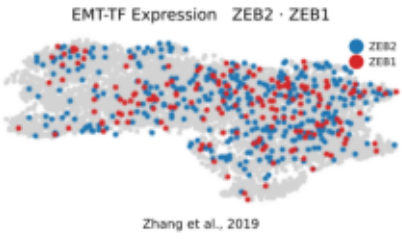
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma / transformed DLBCL (Zhang et al., 2019), 10x — 5,645 malignant B cells over 4 samples (2 patients, each primary + transformed/progressed); Gast-high developmental origin → Crest-high terminal gradient on one continuous manifold.

Verdict.
Pattern reproduces in the non-branching form (Linear): one continuous gastrulation → neural-crest axis, Crest terminal, mild Gast/pSom cross-repression. Gate 1 GO / Gate 2 GO / Gate 3 NA. Gast & Crest confound-independent; pSom proliferation-confounded. EMT-TFs ZEB2/ZEB1 expressed but sparse and not end-polarized.

Parameter choices

Sample threshold	≥100 cells (floor; no dominant cliff, all 4 samples 680–2405) → 4/4 kept, 5,645 cells	**Overdispersed genes**	636 (scFates find_overdispersed); no sc.pp.scale before PCA
Integration	Harmony θ=10 on sample (iLISI peaks at 10, declines at 20/40)	**Embedding**	n_pcs=30, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching scFates curve, N=20 λ=0.3 μ=100; 2 tips/0 forks (SA 2.97, fold-free)	**Root**	curve tip = Gast/pSom-high, Crest-low origin
Module means	Gast +0.012 / pSom +0.006 / Crest +0.009 (weak, control-pool level)	**EMT-TFs**	ZEB2 (27%) · ZEB1 (12%); SNAI/TWIST/FOXC/PRRX near-absent